

Derivatives — group rounds

Lessons 3–4, 7–12 · Quarter I · 40 min · 19 questions · 84 points · groups of 3–4

Four rounds, each timed and worth more than the last. The tag on every question names the topic it comes from. Work on paper, hand in one sheet for the group. An answer with no method scores half.

GROUP

MEMBERS

SCORE

Round A · Easy5'2 pts each

Three topics, eight questions — speed round

1

T2

Evaluate $\lim$ as $x \rightarrow -2$ of $\frac{x^2 + 5x + 6}{x + 2}$.

2

T3

Differentiate $y = 5\sqrt{x} - \frac{3}{x^2}$.

3

T2

Evaluate $\lim$ as $x \rightarrow 3$ of $\frac{x^2 - 9}{x^2 - 2x - 3}$.

4

T2

Evaluate $\lim$ as $x \rightarrow 0$ of $\frac{\sqrt{x+1} - 1}{x}$.

5

T3

Differentiate $y = \frac{2x^3 - 5}{x^2}$.

6

T3

Differentiate $y = x\sqrt{x}$.

7

T4

Differentiate $y = \frac{1}{(2x - 7)^3}$.

8

T4

Differentiate $y = \sqrt[3]{3x + 1}$.

Round B · Medium8'4 pts each

Rewrite before you differentiate

9

T4

Differentiate $y = \sqrt{1 - 4x^3}$.

10

T2

Evaluate $\lim$ as $x \rightarrow 2$ of $\frac{x^3 - 8}{x - 2}$.

11

T2

Evaluate $\lim$ as $x \rightarrow 1$ of $\frac{1}{x - 1} - \frac{2}{x^2 - 1}$.

12

T3

Differentiate $y = (x + 1)(x - 2)(x + 3)$.

13

T3

Differentiate $y = \frac{x^2 + 1}{x^2 - 1}$.

14

T4

Differentiate $y = (1 - 2x)^6(x + 3)$ and factorise.

Round C · Hard9'7 pts each

Two steps at least — be ready to explain

15

T3

Differentiate $y = \frac{x^2 + 1}{x + 2}$ and find every x at which the tangent is horizontal.

16

T2

Evaluate $\lim$ as $x \rightarrow 4$ of $\frac{\sqrt{x} - 2}{x^2 - 16}$.

17

T3

Differentiate $y = \frac{x + 1}{\sqrt{x}}$ for $x > 0$, and give the answer as a single fraction.

18

T4

Differentiate $y = \sqrt{\frac{2x + 1}{2x - 1}}$ for $x > \frac{1}{2}$.

Round D · Very hard8'16 pts each

Every part needed for full marks

19

T3 T4

(a) Show that $y = \frac{x}{\sqrt{(x^2 + 4)}}$ has $y' = \frac{4}{(x^2 + 4)^{3/2}}$. (b) Hence find the equation of the tangent at $x = 0$. (c) Explain why the curve has no stationary point.